Implicit Hierarchical Modeling of Long-Term Structure in Symbolic Music Generation Using xLSTM

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***Abstract*—Deep learning has revolutionized symbolic music generation, yet achieving long-term structural coherence remains a persistent challenge, with generated pieces often lacking thematic development and sectional contrast. Current approaches present a trade-off: implicit architectures, such as standard RNNs and Transformers, excel at local continuation but struggle with global form, while explicitly hierarchical models achieve structural consistency but depend on rigid, pre-defined segmentation that restricts generalization. To address this limitation, this research investigates the Extended Long Short-Term Memory (xLSTM) architecture as a solution for implicit multi-timescale modeling. We hypothesize that xLSTM's novel exponential gating and matrix memory mechanisms allow hierarchical dependencies to emerge organically, enabling the capture of both short-term dynamics and long-term structure without manual segmentation. This study implements an xLSTM-based symbolic generation model and benchmarks it against established baselines, including Music Transformer, MusicVAE, and Museformer. The models are assessed using a comprehensive evaluation strategy incorporating structure-aware metrics, such as motif recurrence and segmentation similarity, alongside human listening tests. Ultimately, this work aims to demonstrate xLSTM's potential to bridge the gap between architectural flexibility and the control required for coherent long-form musical composition.**

***Index Terms*—Symbolic Music Generation, xLSTM, Long-Term Structure, Hierarchical Modeling, structure-aware evaluation**

I. Introduction

Music generation, the use of computational models to create novel musical material has emerged as one of the most dynamic applications of generative artificial intelligence. Beyond artistic exploration, automatic music generation holds practical value in soundtrack production, interactive media, affective computing, and music therapy [1]. Deep learning, in particular, has revolutionized this field by enabling data-driven learning of complex temporal and harmonic patterns from a large corpora of symbolic and audio data. [2], [3]

Despite remarkable progress, long-term structural coherence in generated music remains a persistent challenge [4], [5]. Most deep generative systems excel at capturing short-range dependencies such as motifs, but tend to lose coherence over longer durations. This limitation arises from the difficulty of modeling hierarchical temporal dependencies across multiple timescales spanning motifs, phrases, and sections. [4]

To address this, researchers have proposed explicit hierarchical architectures such as multi-level RNNs, hierarchical LSTMs, hierarchical latent vector models and Transformer variants with segment-level conditioning [6]–[9]. While these designs introduce structural awareness, they depend on pre-defined segmentation of musical form or fixed temporal hierarchies, restricting generalization across genres and compositional styles. Moreover, deeper hierarchies complicate training, causing synchronization issues and exploding parameter counts.

This presents a key **research problem:** *Can we achieve equal or superior long-term structural coherence without relying on manually imposed hierarchical segmentation while still maintaining flexibility, generalizability, and architectural simplicity?*

Addressing this question requires exploring model families capable of learning multi-timescale structures directly from data, without relying on handcrafted structural assumptions. In this research, we examine whether alternative architectures with built-in multi-timescale dynamics can better capture global musical form in symbolic generation tasks.

II. Background

A. Music Theory and Structure

A set of fundamental concepts in music theory used throughout this work is summarized in Table I. These elements represent the core structural components of music.

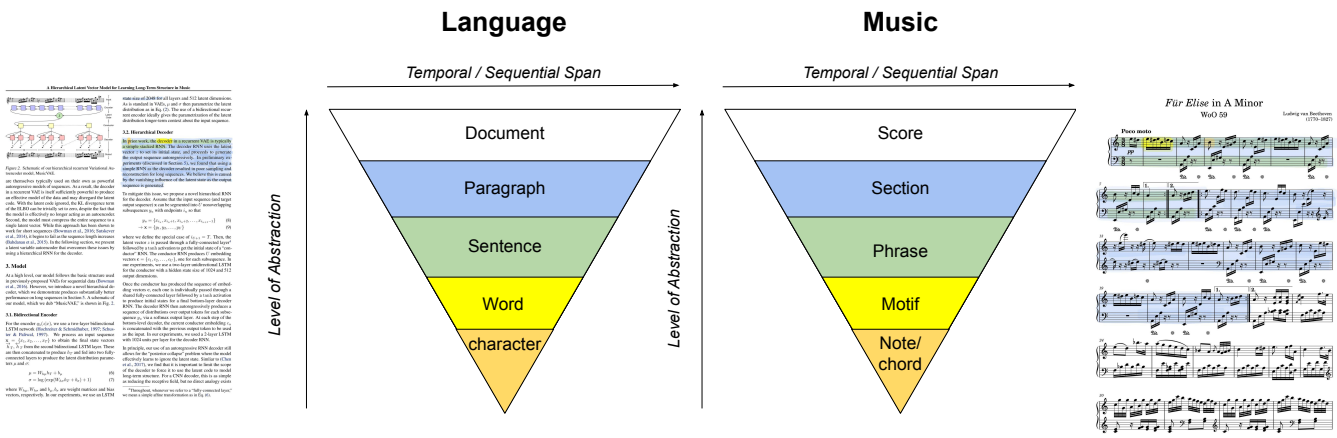

Fig. 1. An Inverted-pyramid representation of hierarchical segmentation in language and symbolic music. Higher layers span longer temporal or sequential units and encode greater levels of abstraction. The conceptual analogy between linguistic and musical structure is inspired by Le et al. [3].

TABLE I
HIERARCHICAL STRUCTURE IN MUSIC

Level	Description
Note	The smallest unit of musical structure, representing a single pitch event.
Motif	A short sequence of notes (typically 2-4) forming a recognizable pattern that recurs throughout a piece.
Phrase	A musical unit comprising multiple motifs (typically 4-8 bars), expressing a complete musical idea analogous to a sentence.
Section	A large-scale structural unit containing multiple phrases, defining formal divisions such as verse, chorus, or theme.

The structural elements of music are organized hierarchically: notes form motifs, motifs combine into phrases, and phrases group into larger sections that realize the global form of a composition. A comparable hierarchy exists in natural language, where characters form words, words grow into sentences, sentences build paragraphs and then the document. This parallelism between musical and linguistic structure provides a useful way to reason about multi-scale dependencies [3]. Figure 1 visualises this shared multi-level organisation using an inverted-pyramid representation, where moving upward corresponds to higher levels of abstraction and moving horizontally corresponds to longer temporal or sequential spans.

B. Symbolic Music Generation Tasks

This paper focuses on **symbolic music**, which, unlike continuous audio waveforms, represents music as a set of discrete, human-readable instructions (e.g., pitch, duration, and onset time). The application of deep learning to symbolic music has expanded the field from simple melody generation to complex, multi-dimensional composition. To navigate this diverse landscape, Ji et al. [10] proposed a comprehensive task-oriented taxonomy which organizes the field into five primary categories.

1) Generation from Scratch: This is the most fundamental form of generation, where the model creates music without external constraints or prior context. It consists of three distinct sub-tasks ranging in complexity:

- **Melody Generation:** Creating a single melodic line with one note sounding at a time (monophonic)
- **Polyphony Generation:** Generating music where multiple notes sound simultaneously within a single instrument or voice.
- **Multitrack Generation:** Generating complete arrangements with multiple independent instruments playing simultaneously, each with its own melodic and harmonic content.

2) Conditional Generation: In this domain, the generation process is guided by specific musical contexts. This is further divided into four sub-tasks:

- **Melody Generation Given Chord:** Generating a melody that fits a provided chord progression.
- **Melody Harmonization:** Generating chord progressions or harmonic accompaniment for a given melody.
- **Accompaniment Arrangement:** Generating multi-instrument backing tracks for a lead melody or chord progression.
- **Score Inpainting:** Filling missing segments within a music score using surrounding musical context.

3) Controllable Generation: This category addresses the need for explicit human control over the generative process by focusing on three high-level attributes:

- **Style:** Controlling the genre or composer style (e.g., style transfer).
- **Structure:** Enforcing global musical forms (e.g., AABA or verse-chorus patterns).
- **Sentiment:** Generating music with specified emotions or valence-arousal values.

4) Performance Generation: While previous tasks focus on the composition, this category focuses on the rendering. It aims to model human-like music characteristics such as micro-timing and dynamics. This is subdivided into *Rendering* (adding expression to an existing static score) and *Composing* (generating expressive performance data directly alongside the notes).

5) Interactive Generation: This category focuses on real-time human-AI co-creation systems. It encompasses *Call and Response* (turn-taking improvisation), *Accompaniment* (real-time backing generation that adapts to human input), and broader *Applications* designed to assist creative workflows.

C. Common Symbolic Music Corpora

The development of deep learning for symbolic music generation has been shaped by the properties of available datasets particularly their scale, quality, and structural annotations [3], [10]. Broadly, existing corpora fall into three categories: (1) large, general-purpose MIDI collections; (2) high-quality performance datasets with expressive detail; and (3) task- or structure-specific datasets tailored to particular aspects of musical modeling.

Large-Scale General MIDI Corpora (Quantity): The single most widely used dataset for large-scale pre-training is the **Lakh MIDI Dataset (LMD)** [11]. It consists of 176,581 unique MIDI files obtained from a large-scale web scrape, intended to be matched against audio databases [11]. As "in-the-wild" data, its quality and metadata are highly variable. A more recent, large-scale dataset is **GiantMIDI-Piano** [12], which contains 10,855 unique classical piano works. This corpus was created by scraping metadata from IMSLP (International Music Score Library Project), finding corresponding audio, and then using a high-resolution piano transcription system to convert the audio into MIDI files [12].

High-Quality Performance Corpora (Quality): In contrast to "in-the-wild" data, a second category of datasets focuses on high-fidelity, expressive performances. The most prominent is the **MAESTRO** dataset [13], containing over 172 hours of piano performances. Its defining feature is the fine alignment (≈ 3 ms) between note labels and audio waveforms [13], which made it the foundational dataset for expressive performance models like the Music Transformer [14]. Similarly, the **ASAP (Aligned Scores and Performances)** dataset provides 1,068 performances aligned to 222 digital scores (in MusicXML and quantized MIDI) [15]. It was explicitly designed to support score-to-performance research. An earlier, related effort was the **MusicNet** dataset, which provided 34 hours of classical music by aligning MIDI scores to audio recordings using Dynamic Time Warping (DTW) [16].

Task-Specific & Structurally-Specific Corpora: These datasets have been curated for specialized research tasks:

- **For Arrangement:** The **POP909** dataset [17] contains 909 popular songs with separate, aligned MIDI tracks for vocal melody, lead instrument, and piano accompaniment.
- **For Drumming:** The **Groove MIDI Dataset (GMD)** [18] provides 13.6 hours of tempo-aligned expressive drumming from electronic drums, with fine-grained timing and dynamics for groove modeling.
- **For Emotion:** The **EMOPIA** dataset [19] contains 1,087 pop piano clips with emotion labels based on Russell's circumplex model. It is designed for emotion recognition and conditioned generation.
- **For Harmony/Score:** The **JSB Chorales** dataset [20] provides 389 four-part Bach chorale harmonizations. Its clean, homogeneous structure makes it a standard benchmark for harmonic modeling, as used in DeepBach. [20].

As summarized in Table II, most large-scale corpora used for training deep generative models remain fundamentally tied to the **MIDI** format due to the wide availability of MIDI data in the music industry. This dominance has shaped the field's modeling assumptions and highlights the need to critically examine the limitations of MIDI as a symbolic format, discussed in the next subsection.

D. MIDI & The Symbolic Music Representation Problem

MIDI itself was never designed as a score representation but rather it is a real-time messaging protocol to control synthesizers, introduced in the early 1980s [21].

In a MIDI stream, a single, semantically whole musical concept such as "a C4 note, played forte for one beat" is not a single unit. Instead, it is fragmented into a sequence of separate, time-ordered events, such as a `NOTE_ON` event (pitch C4), a `SET_VELOCITY` event (forte), and, much later, a `NOTE_OFF` event (stop C4). This mismatch between musical concepts and their serial representation forces an artificial sequentiality onto the data [3], [22].

The central challenge lies in determining how to organise and reshape this fragmented flow of musical events so that it can be processed effectively by sequence models, an issue examined in the following sections.

III. Literature Review

AI-based symbolic music generation has evolved through advances in data representation, model architectures, and evaluation methodologies. Early rule-based and probabilistic systems relied on explicitly encoded musical rules, while deep learning methods enabled data-driven discovery of temporal and harmonic patterns. These advances have enhanced the stylistic and local coherence of generated music. However, capturing *long-term musical structure*—including repetition, motifs and sectional organization remains an open challenge that depends on how music is represented, modeled, and evaluated [2]–[5].

To provide a comprehensive understanding of how these dimensions contribute to long-term coherence structural modeling, this review adopts a three-axis framework encompassing:

- **Representation Axis** – how musical information is encoded and tokenized for machine learning models, including symbolic encodings (MIDI, REMI, CP) and data compression strategies (e.g., BPE). Representation choices determine how much structural context (bar, phrase, or section) is accessible to the model.

TABLE II
A SUMMARY OF COMMON SYMBOLIC MUSIC CORPORA

Corpus	Format / Type	Scale	Key Characteristic / Primary Use
Lakh MIDI (LMD) [11]	General MIDI	176,581 files	"In-the-wild" quantity. Large-scale pre-training.
GiantMIDI-Piano [12]	Transcribed MIDI	10,855 works	Large-scale classical piano, transcribed from audio.
MAESTRO [13]	Performance MIDI + Audio	172 hours	High-quality, expressive virtuosic piano performances.
ASAP [15]	Score (MusicXML) + Perf. MIDI	1,068 perfs.	Explicit, fine-grained score-to-performance alignment.
MusicNet [16]	Aligned Score + Audio	34 hours	Aligned classical music for note prediction (DTW-based).
POP909 [17]	Structured MIDI (Multi-track)	909 songs	Pop songs with separate melody/accompaniment/chord tracks.
Groove MIDI (GMD) [18]	Performance MIDI (Drums)	13.6 hours	Expressive drum performances with micro-timing/velocity.
EMOPIA [19]	Performance MIDI + Audio	1,087 clips	Pop piano clips with human-annotated emotion labels.
JSB Chorales [20]	Score (4-part)	389 chorales	Small, clean, homogeneous corpus for harmonic modeling.

- **Architecture Axis** – the evolution of model families used for music generation, from recurrent and convolutional networks to attention-based and hierarchical architectures.
- **Evaluation Axis** – the methodologies and metrics used to assess musical coherence and structure, including both quantitative measures (e.g., structural similarity, phrase consistency) and perceptual human evaluations.

Together, these three perspectives provide a comprehensive understanding of the field's progression and highlight the motivation for exploring implicit multi-timescale modeling using xLSTM.

A. The Representation Axis: Encoding Music for Long-Term Coherence

1) Introduction: Why Representation is the Key: Generating coherent long-term musical structure fundamentally depends on representation. Before a model can learn the high-level relationships that define a composition, such as the repetition of a theme or a harmonic progression across sections, the music must be converted into a sequential format that the model can ingest. This choice of representation presents a core trade-off: it must be musically expressive enough to capture the nuances of pitch, rhythm, and harmony, yet simultaneously be efficient enough for a sequence model to learn dependencies over thousands of time steps.

As noted in recent surveys [3], [10], the evolution of symbolic music generation, particularly in the context of Transformer models, is fundamentally linked to the evolution of these representations. This section will trace this evolution, arguing that the critical shift from dense, time-slice-based representations to sparse, event-based representations and then from *elementary* event tokens to *composite* "compound word" tokens enabled models to learn coherent long-term structure.

2) The Core Dichotomy: Time-Slice vs. Event-Based Tokenization: At the highest level, symbolic music representations are divided by their core philosophy for handling time, a categorization highlighted by [3], [10].This fundamental split creates a hierarchical taxonomy, which is visualized in Figure 2. The two primary branches are:

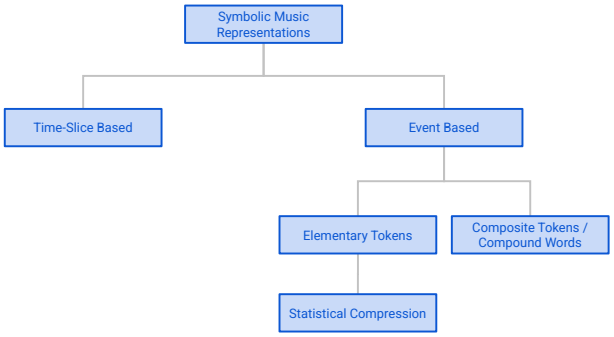

Fig. 2. A hierarchical taxonomy of symbolic music representations. The primary branch separates Time-Slice Based methods from Event-Based methods. The Event-Based branch is then divided based on token definition: Elementary Tokens, which can be modified by Statistical Compression (e.g., BPE), and Composite "Compound Word" Tokens.

1) **Time-Slice Representations:** These methods discretize time into a constant grid (e.g., 16th notes). The token at each step is a vector representing all musical information active during that slice. This "matrix-like" or "image-like" approach [23] is exemplified by the widely-used piano roll.
2) **Event-Based Representations:** These methods discretize time by *musical events*. The sequence only advances when a new event occurs, such as a note being played or a rest beginning.

As Figure 2 illustrates, the Event-Based approach splits based on token definition. This crucial split—between **Elementary Tokens** and **Composite Tokens**—and the subsequent development of **Statistical Compression** methods (like BPE) to solve the flaws of the elementary approach, forms the core of the following analysis.

3) Time-Slice Representations: The earliest and most intuitive methods for sequential modeling, time-slice representations, are exemplified by piano rolls and their variants.

While intuitive for image-based models like CNNs [23], this approach has two fundamental flaws that make it problematic for modeling long-term structure:

1) **Sequence Length Explosion:** A single bar of 4/4 music at 16th-note resolution generates 16 tokens (or steps) *per voice*. A minute-long piece can easily become thousands of steps long, exceeding the context window of most sequence models.
2) **Inefficient Rhythm Representation:** A quarter note, for example, is not a single token but is represented implicitly as four identical, consecutive time-step tokens [20]. As noted by Ji et al. [10], this makes it difficult for a model to learn rhythmic patterns.

These limitations make time-slice representations impractical for long-form generation.

4) Event-Based Representations: Event-based representations solve the sequence-length problem by creating a sparser sequence where tokens only represent a *change* in the musical state. This approach, however, introduces a new, critical design choice: what, precisely, constitutes a single event token? To answer this, we trace how the field evolved from elementary tokens to compound tokens, examining key refinements along the way.

a) Elementary / Individual Tokens (NLP-like Approach): This approach defines each token as a single, discrete musical atom [10] [3]. The 'Music Transformer' [14] exemplifies this, using a vocabulary of `NOTE_ON` pitches, `NOTE_OFF` pitches, `SET_VELOCITY` tokens, and `TIME_SHIFT` tokens.

While this representation is highly expressive and capable of capturing subtle performance timings, it creates two new problems for modeling long-term structure:

1) **Long Sequences:** The sequences, while sparser than time-slice, are still extremely long. A single note requires at least four tokens (`SET_VELOCITY`, `NOTE_ON`, `TIME_SHIFT`, `NOTE_OFF`). The *entire* architectural innovation of the Music Transformer—Relative Attention was created specifically to handle the thousands of steps that this representation generates for just a minute-long piece [14].
2) **Artificial Sequentiality:** As critiqued by Le et al. [3], splitting a single musical concept (a C4 quarter note) into an *arbitrary* sequence of tokens forces the model to learn a *sub-grammar* (e.g., that a `NOTE_ON` must eventually be followed by a `NOTE_OFF` of the same pitch) before it can learn the *musical* grammar. This complex nested structure [22] can lead to ill-formed outputs, such as dangling notes that never turn off.

b) Metrically-Informed Elementary Tokens: REMI: Before the transition to composite tokens, an important refinement to the elementary event-based approach was introduced by **REMI** [24]. REMI remains an *elementary* representation as each musical note is still decomposed into multiple tokens. However, REMI significantly improves rhythmic modeling by replacing fine-grained `TIME_SHIFT` tokens with explicit *metrical markers* such as `BAR` and `POSITION`. This makes the musical grid explicit and greatly helps Transformers learn downbeat structure and phrase alignments.

c) A Statistical Solution: Byte-Pair Encoding (BPE): Before the field fully adopted the semantic grouping of compound words, one proposed solution to the sequence length problem was to apply subword tokenization techniques, most notably Byte-Pair Encoding (BPE), borrowed directly from NLP [3], [10]. BPE is a compression technique [23] that iteratively merges the most frequent pairs of tokens in a corpus into a single new token.

In theory, this approach could learn to merge a `NOTE_ON` event and its corresponding `DURATION` into a single, shorter token. However, empirical studies report limited gains. Zhang et al. [23] observed no meaningful improvement in classification performance when applying BPE to symbolic music. Likewise, Le et al. [3] note that although BPE produces a more structured embedding space, it does not translate into noticeable performance benefits on music analysis tasks. These findings suggest that purely statistical compression is insufficient; instead, improving symbolic representations likely requires a *semantically* informed redesign of the token itself

d) The Solution for Structure: Compound Word / Composite Tokens: This second approach, identified as "Compound Word" [10] or "Composite Token" [3], provides a direct solution to the problems of the elementary approach. A compound word groups all the attributes of a single musical event (one note) into a *single, atomic token* or "super-token."

This shift in representation has profound implications for modeling long-term structure. First, it **eliminates artificial sequentiality** entirely. The token `(Pitch=60, Duration=0.5)` is a single unit; the model no longer learns the *order of attributes* and can focus exclusively on the *sequence of musical events*. Second, it results in **massive sequence compression**. A bar of music that required hundreds of elementary tokens may now be represented by only a few dozen compound tokens.

This evolution is clearly documented in a series of state-of-the-art papers:

- **CPWord (Compound Word Transformer):** Hsiao et al. [25] introduce the **Compound Word (CPWord)** representation, which groups related musical attributes into a single "family" (e.g., note, metric) and predicts them jointly within one time step. This design substantially shortens token sequences, allowing the model to operate more efficiently and making full-song symbolic composition more feasible.
- **MuMIDI (PopMAG):** Ren et al. [26] propose **MuMIDI**, a representation for multi-track symbolic generation that encodes all note attributes (pitch, duration, velocity) within a single prediction step. This consolidation shortens the effective sequence length and enables the model to better capture long-range musical structure.

TABLE III
COMPARATIVE ANALYSIS OF KEY SYMBOLIC MUSIC REPRESENTATIONS

Representation (Paper)	Type	Core Unit	Rhythm Handling	Relative Sequence Length	Implications for Long-Term Structure Modeling
DeepBach [20]	Time-slice	Fixed timestep (e.g., 16th note)	Implicit ("hold" tokens)	Very long (e.g., 16 steps/bar/voice)	Sequence length scales linearly with temporal resolution, effective long-range context.
Music Transformer [14]	Event-based (elementary)	Musical attribute (e.g., `NOTE_ON`)	Explicit (`TIME_SHIFT` tokens)	Long (one note = 3–4+ tokens)	Expressive timing supported, but long event chains per note hinder global structure.
BPE-Tokenized [23]	Event-based (compressed)	Statistically merged tokens	Implicit (merged by frequency)	Compressed	Improved compactness but limited gains in structure modeling.
REMI [24]	Event-based (elementary)	Musical event (note/metric)	Explicit & metrical (`BAR`, `POSITION`)	Long	Explicit bar and position tokens support metrical and phrase-level organization.
OctupleMIDI [27]	Event-based (composite)	Musical note as 8-tuple	Explicit & metrical (within tuple)	Highly compressed ($\approx 4\times$ shorter than REMI)	Note-level composite tokens greatly reduce sequence length and improve global coherence.

- **OctupleMIDI (MusicBERT):** Zeng et al. [27] propose **OctupleMIDI** as a response to the excessive sequence lengths of REMI-style representations, which average over 15,000 tokens per piece. OctupleMIDI encodes each note as a compact, note-centric 8-tuple, reducing the sequence length by roughly a factor of four compared to REMI. This substantial compression makes the representation far more suitable for large-scale pre-training.

5) Critical Evaluation and Conclusion: The progression from dense time-slice representations to event-based and compound-word encodings reflects a shared conclusion across the literature: *long-term musical structure cannot be learned effectively at the level of isolated, low-level events*. Instead, successful long-form generation depends on compressing fine-grained attributes into higher-level composite tokens that align more closely with musically meaningful units such as notes, motifs, phrases, or bars. By reducing effective sequence length and increasing semantic density per token, models are better positioned to reason over long-range dependencies rather than being dominated by intra-note details.

Table III summarizes how major symbolic representations operationalize this principle. While compound-word and metrical event representations have been instrumental in enabling long-form generation, they introduce several non-trivial trade-offs that constrain their general applicability.

The most significant trade-off is **vocabulary size**. Elementary token vocabularies (e.g., Music Transformer) typically contain only a few hundred tokens. In contrast, compound-word vocabularies must encode entire combinations of musical attributes, resulting in substantially larger embedding spaces. Both Hsiao et al. [25] and Ren et al. [26] note that such vocabularies increase the parameter count of the embedding layers and require larger datasets for effective training.

The second trade-off is **expressive timing**. Compound-word formats such as MuMIDI and OctupleMIDI are inherently tied to a fixed metrical grid (e.g., 16th-note quantization), which simplifies score-based structure but reduces the capacity to represent fine-grained timing nuances such as rubato or swing. Huang et al. [24] highlight that these representations, while advantageous for modeling meter and harmony, sacrifice the temporal flexibility afforded by event-based encodings like the `TIME_SHIFT` tokens used in Music Transformer [14].

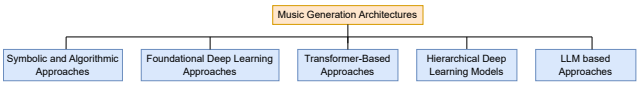

Fig. 3. Taxonomy of music generation architectures showing the evolution from rule-based systems to implicit multi-timescale models.

Finally, representation choice remains **task-dependent**. While compound-word formats consistently benefit long-form generative modeling, comparative studies indicate that matrix-like representations remain competitive for classification and analysis tasks, offering improved robustness and memory efficiency [23]. This underscores that no single representation is universally optimal across symbolic music processing.

B. Architectures

Figure 3 presents a high-level taxonomy of music generation architectures, illustrating the chronological evolution from symbolic and algorithmic approaches through deep learning methods to modern hierarchical and implicit multi-timescale models. This taxonomy provides a roadmap for understanding how the field has progressively addressed the challenge of long-term structural coherence in music generation.

1) Symbolic and Algorithmic Approaches: Early symbolic systems tried to model music through explicit rules or probabilistic transitions between notes. One of the most influential examples is the Variable-Length Markov Model with Reinforcement Learning proposed by Cont et al. (2006) [28]. In this system, multiple agents learned musical attributes such as pitch, rhythm, and harmony through a reward-based process. The model could recognize phrase boundaries and repeat short motifs by using Factor Oracles which are data structures that store repeating patterns efficiently. This approach produced

TABLE IV
SUMMARY OF SYMBOLIC AND ALGORITHMIC APPROACHES

Approach	Core Technique	Strength	Limitation
Cont et al. (2006) [28]	Variable-Length Markov + RL, Factor Oracles	Phrase recognition	Predefined rules, no generalization
Pachet & Roy (2011) [29]	Constraint programming Markov	Explicit control	Manual constraints, limited creativity
Collins & Laney (2017) [30]	Pattern discovery (SIACT), template-based	Rhythm consistency	Lacks originality
Herremans et al. (2015, 2017) [31], [32]	Variable Neighborhood Search	Cyclic patterns, tonal tension	Tuning complexity
Alvarado et al.; Ting et al. [33], [34]	Genetic Algorithms	Structured evolution	Fitness tuning, scalability

TABLE V
SUMMARY OF FOUNDATIONAL DEEP LEARNING APPROACHES

Approach	Core Technique	Strength	Limitation
Lookback RNN (2016) [35]	RNN with lookback inputs (1-2 bars)	Notices repetition, maintains short motifs	Limited larger forms (verse-chorus)
Attention RNN (2016) [35]	RNN with early self-attention	Looks back at any previous note	Limited sustaining of larger forms
StructureNet (2018) [36]	Auxiliary network for pattern recognition	Consistent repetition, phrase boundaries	Confined to local structures
WaveNet (2016) [37], [38]	Dilated convolutions	Large temporal view, long-term repetition	Struggles with hierarchical forms

compositions with short sections and phrase-like divisions, but it depended on predefined rules and could not generalize to new musical styles.

To overcome some of these limitations, Pachet and Roy (2011) [29] introduced a constraint programming method for generating Markov sequences under user-specified rules. Instead of learning from data, this system searched for note sequences that satisfied both probabilistic and structural constraints like repeating motifs, proper endings, and phrase balance. The advantage was explicit control over structure, but it required the user to define many musical conditions manually, limiting creativity and adaptability.

Another key system, Racchmaninof-Jun2015 by Collins and Laney (2017) [30], aimed to inherit repetitive patterns from an existing "template" composition. Using a pattern discovery algorithm (SIACT), the model identified recurring 2-bar motifs from the template and then generated new melodies that followed similar repetition structures. Its improved "beat-relative" representation of MIDI data helped it maintain consistent rhythm, but the generated music often stayed too close to the template, lacking originality.

Optimization-based methods such as Herremans et al. (2015) [31] and Herremans & Chew (2017) [32] used Variable Neighborhood Search (VNS) and pattern extraction to guide melodic creation. These models were able to maintain global cyclic patterns and tonal tension, especially in genres like traditional Bagana or polyphonic music. Similarly, Genetic Algorithms (Alvarado et al., 2020 [33]; Ting et al., 2015 [34]) evolved melodic phrases by mutating and recombining short motifs. While these approaches produced music with noticeable structure, they required carefully tuned fitness functions and could not easily scale to complex datasets.

Table IV summarizes the key symbolic and algorithmic approaches, highlighting their core techniques, strengths, and limitations.

2) Foundational Deep Learning Approaches: The next phase of progress came with deep learning, which replaced rule-based control with data-driven learning. Instead of defining rules about when to repeat or modulate, neural networks learned these patterns from thousands of examples.

Two early models from Google Magenta (Waite et al., 2016) [35] Lookback RNN and Attention RNN are prominent work to apply Recurrent Neural Networks (RNNs) to melody generation. Lookback RNN gave the network "lookback" inputs, showing it what notes were played 1 or 2 bars earlier. This helped the model notice repetition and maintain short motifs. Attention RNN added an early form of self-attention, letting the network "look back" at any previous note rather than relying entirely on memory. Both models improved short-term coherence, but their ability to sustain larger forms (like verse–chorus patterns) was still limited.

Building on this, StructureNet (Medeot et. al., 2018) [36] introduced an auxiliary network trained to recognize structural patterns, such as repetition, from real melodies. During generation, StructureNet worked alongside the main RNN to bias note selection toward repeated sequences seen in training. The result was music with more consistent repetition and phrase boundaries, though still confined to local structures.

Another notable advance was WaveNet (Van den Oord et al., 2016) [37], later adapted for symbolic music [38]. Unlike RNNs, WaveNet used dilated convolutions, allowing the model to see much farther back in time by skipping over notes at increasing intervals. This design gave the model a large "view" over the melody, enabling it to generate rhythms and patterns that stayed consistent across many bars. WaveNet outperformed LSTMs in capturing long-term repetition, but convolutional filters still struggled with truly hierarchical forms like sections or themes.

Table V summarizes the foundational deep learning approaches and their contributions to modeling musical structure.

3) Transformer-Based Approaches: The introduction of Transformers revolutionized music generation. Instead of processing sequences one step at a time, Transformers use self-

attention to relate every note to every other note in the sequence, enabling powerful long-range modeling.

The Music Transformer (Huang et al., 2019) [14] applied this architecture to MIDI music, using relative positional encoding to preserve timing relationships. It could generate long, stylistically rich piano performances that maintained rhythmic consistency. However, it still lacked explicit understanding of form or repetition as it could mimic style but not large-scale musical structure.

The Pop Music Transformer (Huang & Yang, 2020) [24] introduced REMI (Revamped MIDI Events), a new representation that encodes bars, beats, tempo, and chords. By giving the model a "musical grid" to follow, it learned rhythm and harmony more naturally. Objective and listening tests showed that REMI produced clearer downbeats and smoother timing than standard MIDI representations. The only drawback was that encoding all this extra information made input sequences very long, which limited scalability for full-length pieces.

To address the issue of lost themes, the Theme Transformer (Dai et al., 2021) [39] focused on thematic continuity. It automatically discovered short recurring melodic phrases using contrastive learning and clustering, then conditioned the Transformer on these themes through cross-attention. The model successfully repeated themes across its output, achieving a more human-like sense of recurrence. However, it could not yet develop those themes and it repeated them too literally without creative variation.

Other works explored structural awareness from different angles. The R-Transformer (Hu et al., 2022) [40] modeled repetition types directly, distinguishing between exact repeats and variations. The Museformer (Yu et al., 2022) [41] introduced hierarchical attention layers, allowing the model to process music at multiple time scales (notes → phrases → sections). These systems brought deep learning closer to capturing musical form, but their attention windows and training complexity still limited how much structure they could represent over long durations.

Table VI summarizes the key Transformer-based approaches and their advancements in modeling musical structure.

TABLE VI
SUMMARY OF TRANSFORMER-BASED APPROACHES

Approach	**Core Technique**	**Strength**	**Limitation**
Music Transformer (2019) [14]	Self-attention with relative positional encoding	Long, stylistically rich performances	Lacks explicit form understanding
Pop Music Transformer (2020) [24]	REMI representation (bars, beats, tempo, chords)	Natural rhythm and harmony learning	Long sequences limit scalability
Theme Transformer (2021) [39]	Contrastive learning + cross-attention on themes	Human-like thematic recurrence	Literal repetition, no theme development
R-Transformer (2022) [40]	Models repetition types (exact vs. variations)	Distinguishes repeat patterns	Limited by attention windows
Museformer (2022) [41]	Hierarchical attention layers	Multi-scale processing (notes → sections)	Training complexity

TABLE VII
SUMMARY OF LLM-BASED ARCHITECTURES FOR SYMBOLIC MUSIC GENERATION

Approach	**Core Technique**	**Strength**	**Limitation**
LakhNES (2019) [42]	Transformer-XL with NES-specific tokenization	Arbitrarily long generation in constrained domain	Rigid four-voice format; limited generalization
MuPT (2024) [43]	Large-scale decoder-only Transformer pre-trained on ABC / SMT-ABC	Strong continuation quality via massive pre-training	High computational cost; limited conditioning flexibility
ChatMusician (2024) [44]	Instruction-tuned LLM with symbolic music as a second language	Unified text reasoning and symbolic music generation	Genre bias, hallucinations, and memorization risk

4) Large Language Model–Based Architectures: A second line of work treats symbolic music directly as a sequence of discrete tokens and applies large language model (LLM)–style architectures. These systems rely on autoregressive Transformer or Transformer-XL decoders trained with next-token prediction, distinguishing themselves mainly through representational choices, the scale of pre-training, and whether they specialize in music or share capacity with natural language tasks.

a) LakhNES: early Transformer-XL for multi-instrumental scores: LakhNES adapts Transformer-XL to generate four-voice NES-style chiptune music, representing each event using an NES-specific vocabulary. [42] The model generates arbitrarily long continuations, fills missing melodic material, and benefits from a cross-domain pre-training strategy in which the heterogeneous Lakh MIDI corpus is mapped into the fixed NES four-voice format. This pre-training yields about a 10% relative improvement and higher listener preference. [42] However, the system remains tied to a rigid four-instrument domain and does not target general symbolic music or open-ended stylistic variation.

b) MuPT: foundation-style symbolic music Transformers: MuPT advances toward a foundation model for symbolic music by pre-training decoder-only Transformers (190M–4.23B parameters) on 33.6B tokens encoded in ABC notation. [43] The authors motivate ABC as a compact, text-like format and introduce Synchronized Multi-Track ABC (SMT-ABC) to maintain track alignment during generation. [43] Larger MuPT variants with 8192-token contexts cover the majority of

training sequences and show strong scaling behavior: human listeners significantly prefer MuPT completions over GPT-4 and a prior symbolic model (MMT), and the model reaches quality comparable to "mediocre human composers." [43] Despite its scale, MuPT remains focused on continuation-style tasks and does not incorporate natural-language conditioning beyond simple prompts.

c) ChatMusician: general LLM with intrinsic music abilities: ChatMusician takes the opposite approach by extending a LLaMA2-based LLM with intrinsic symbolic music proficiency. [44] Using 4B tokens from the MusicPile corpus and instruction-style fine-tuning, the model treats music as a "second language" and unifies theory question answering, text-conditioned composition, ABC score generation, and multi-modal chat. [44] It outperforms GPT-4 and GPT-3.5 across symbolic generation and music-theory benchmarks while maintaining general language ability. The authors also report limitations: the model exhibits strong genre bias (particularly toward Irish/folk styles), occasional hallucinations in open-ended compositions, and measurable memorization risk. [44]

d) Comparison and critical discussion: Together, these three systems trace a trajectory from domain-specific Transformers to billion-parameter, instruction-following LLMs for symbolic music. LakhNES demonstrates that Transformer-XL can generate coherent long sequences within a tightly constrained domain. MuPT scales this approach dramatically, achieving strong continuation quality via massive pre-training on ABC/SMT-ABC. ChatMusician broadens the scope by coupling text-based reasoning with symbolic generation, though at the cost of genre bias and instability.

Architecturally, however, all three models share a common property: they operate as *flat, autoregressive Transformer decoders* with no mechanism—implicit or explicit—for multi-timescale or hierarchical structure. Their long-term coherence arises largely from scale (longer context windows, larger pre-training corpora), rather than from architectural inductive biases toward hierarchical musical form. Moreover, each system faces documented challenges: domain rigidity (LakhNES) [42], heavy computational demands and limited conditioning flexibility (MuPT) [43], and dataset-induced biases and hallucinations (ChatMusician) [44].

Table VII summarizes the key LLM-based approaches and their advancements in modeling musical structure.

5) Explicit Hierarchical Deep Learning Models: Music is inherently hierarchical: motifs form phrases, phrases form sections, and sections together create a complete composition. Hierarchical deep learning models have become central to advancing automatic music generation, particularly for capturing the long-term structure that characterizes human-composed music. Traditional deep learning models, such as standard RNNs and LSTMs, often struggle to generate music with coherent global structure, leading to outputs that lack musicality and organization over extended timescales. In response, researchers have developed hierarchical architectures spanning RNNs, Transformers, VAEs, GANs, and hybrid models that explicitly model music at multiple temporal and structural levels (e.g., notes, bars, phrases, sections) [6], [8], [45]–[47]

a) Whole-Song Hierarchical Generation of Symbolic Music: A recent work on Whole-Song Hierarchical Generation of Symbolic Music [48] introduces a four-level hierarchical framework for full-song symbolic music generation using cascaded diffusion models. Addressing the challenge of maintaining coherence in long compositions, the authors define a computational hierarchical music language comprising four layers form, reduced lead sheet, lead sheet, and accompaniment each modeling progressively finer musical detail. Music is represented as multi-channel piano-roll images and generated in a top-down manner, where each diffusion model conditions on the higher-level output to ensure consistency across sections. The framework achieves efficient and controllable whole-song generation, exhibiting clear verse–chorus structures and high phrase-level consistency. To evaluate structural quality, the study proposes the Inter-Phrase Latent Similarity (ILS) metric, which quantifies phrase coherence using latent representations from pre-trained VAEs, alongside listener-based assessments of boundary clarity, phrase similarity, naturalness, and musicality. Experimental results demonstrate that this hierarchical diffusion approach outperforms baseline models in both structural coherence and perceptual quality, highlighting the effectiveness of explicit hierarchical modeling and cascaded control for capturing long-term musical structure.

b) Small Tunes Transformer (STT): The Small Tunes Transformer [9] proposes a hierarchical Transformer framework for melody generation conditioned on structural skeletons, focusing on traditional Chinese folk music. Addressing the lack of deep learning research on non-Western musical forms, the model captures both macro-level (phrase segmentation) and micro-level (skeleton notes) hierarchies to preserve thematic and rhythmic coherence. A novel Phrase-level Cross-Attention mechanism allows the decoder to attend only to skeleton notes within the same phrase, effectively linking phrase-scale and note-scale dependencies. The study also introduces the Small Tunes Dataset of over 10,000 MIDI files and three objective metrics Theme Pitch Correspondence (TPC), Theme Rhythm Correspondence (TRC), and Pentatonic Scale Consistency (PSC) to evaluate pitch, rhythm, and style coherence. Experimental results show that STT produces more musically coherent and stylistically faithful melodies than baseline models, confirming that explicit hierarchical modeling and phrase-level conditioning substantially improve long-term structural consistency in symbolic music generation.

c) Hierarchical Multi-Head Attention LSTM for Polyphonic Symbolic Melody Generation: Authors propose a Hierarchical Multi-Head Attention LSTM [7] model to address the difficulty of generating polyphonic symbolic melodies that maintain musical coherence and long-term dependency. Conventional RNN-based methods often fail to capture the rich inter-dependencies among musical events, resulting in repetitive or incoherent outputs. The proposed model integrates hierarchical representations with multi-head attention to learn both local and global dependencies across different abstrac-

tion levels, enabling the generation of expressive, structurally consistent melodies. Trained and evaluated on the MAESTRO and Classical-Music MIDI datasets, the model achieved lower mean-squared-error losses and produced compositions judged by listeners as more coherent and musically engaging on Likert-scale evaluations. These findings highlight that combining hierarchical recurrence with attention mechanisms enhances the modeling of long-term structure and polyphonic texture in symbolic music generation.

d) PiRhDy: The PiRhDy [49] framework introduces a hierarchical embedding model that jointly learns representations of pitch, rhythm, and dynamics to capture the complex nature of symbolic music. Unlike traditional word embeddings, which treat musical events as single tokens, PiRhDy models each note as a combination of chroma, octave, inter-onset interval, note state, and velocity, thereby integrating melodic and harmonic dimensions. Its two-stage hierarchical design first constructs token-level embeddings from these features and then refines them using context modeling that incorporates both horizontal (melodic) and vertical (harmonic) dependencies through RNN and attention layers. Extensive experiments demonstrate that this integrated representation enhances performance across melody completion, accompaniment generation, and genre classification tasks. The study's findings confirm that multi-level hierarchical embeddings capturing pitch, rhythm, and dynamics interactions provide a richer and more musically grounded latent space, advancing structure-aware symbolic music modeling.

e) MusicVAE: Authors introduced MusicVAE [8], a hierarchical variational autoencoder (VAE) designed to overcome the limitations of recurrent VAEs in modeling long-term structure in musical sequences. The model employs a two-level hierarchical decoder, where a "conductor" RNN first generates embeddings for subsequences (e.g., musical bars), which then guide a lower-level decoder RNN to produce fine-grained note events. This design effectively mitigates posterior collapse and extends the VAE's ability to represent long-horizon temporal dependencies. Experimental results on large-scale MIDI datasets show that the hierarchical decoder yields substantially higher reconstruction accuracy and smoother latent-space interpolations than flat RNN-based baselines. Furthermore, human listening tests revealed that samples from the hierarchical model were rated as more musical and coherent, comparable even to real data. Critically, while the model achieves notable improvements in structural coherence and creative controllability, it still relies on explicit hierarchical segmentation (fixed-length bar divisions) and a static latent representation, limiting its adaptability to variable structural boundaries or complex global forms.

Table VIII summarizes the hierarchical deep learning models and their approaches to capturing long-term musical structure.

Collectively, these hierarchical deep learning approaches demonstrate the effectiveness of explicit hierarchy construction in modeling long-term musical dependencies. Models such as MusicVAE and STT highlight the benefits of separating phrase- and note-level processing, while hierarchical diffusion and attention-based systems show that multi-scale conditioning enhances structure and controllability. However, these models depend on predefined segmentation schemes (bars, phrases, or sections), requiring hand-crafted hierarchies and rigid synchronization between layers. Such reliance on explicit boundaries constrains their adaptability to irregular musical forms and dynamic temporal variations. This persistent limitation motivates the exploration of architectures like xLSTM, which introduce implicit hierarchical memory through multi-timescale gating mechanisms, potentially enabling models to learn long-term structure organically without explicit hierarchical design.

TABLE VIII
SUMMARY OF EXPLICIT HIERARCHICAL DEEP LEARNING MODELS

Approach	**Core Technique**	**Strength**	**Limitation**
Whole-Song Hierarchical (2024) [48]	4-level cascaded diffusion models	Clear verse-chorus structures, phrase consis-tency	Requires explicit hierarchical design
Small Tunes Transformer (2024) [9]	Phrase-level cross-attention on skeletons	Thematic and rhythmic coherence	Predefined phrase segmentation
H-Multi-Head LSTM (2024) [7]	Hierarchical LSTM with multi-head attention	Local and global dependen-cies	Fixed abstraction levels
PiRhDy (2020) [49]	Two-stage hierarchical embedding (pitch, rhythm, dynamics)	Richer latent space	Requires horizontal and vertical dependency modeling
MusicVAE (2018) [8]	Conductor-decoder VAE hierarchy	Mitigates posterior collapse	Fixed-length bar divisions, static latent

6) xLSTM: An Implicit Multi-Timescale Alternative: The Extended Long Short-Term Memory (xLSTM) architecture revisits recurrent sequence modeling through two core innovations: exponential gating and matrix memory (mLSTM) which together enable implicit multi-timescale dynamics within a single recurrent layer [50]

a) Exponential Gating and Memory Mixing: xLSTM introduces exponential gating and memory mixing, which allow for more effective storage decisions and the handling of rare events or long-term dependencies. The gating mechanism dynamically selects which information to retain or forget, enabling the model to adaptively focus on relevant temporal features at different scales [51]–[53]

b) Matrix Memory and Parallelization: The matrix LSTM (mLSTM) variant of xLSTM employs a matrix-valued memory cell and covariance update rules, significantly increasing memory capacity and allowing for efficient, parallel updates. This structure is particularly suited for modeling relationships across multiple time scales and capturing hierarchical dependencies in sequential data. [51], [52]

c) Empirical Performance: While direct evaluations of xLSTM architectures in music generation are not yet available, cross-domain evidence strongly supports their effectiveness in modeling long-term dependencies. Across tasks demanding extended temporal reasoning such as time-series forecasting, wind energy prediction, and lip reading, xLSTM-based models consistently outperform both traditional LSTM and Transformer counterparts. These improvements are attributed to xLSTM's enhanced memory retention and implicit multi-timescale dynamics, which enable the network to model hierarchical temporal structure more effectively. Notably, the performance advantage becomes more pronounced as sequence length or prediction horizon increases, underscoring xLSTM's superior ability to capture long-term and hierarchically organized patterns in sequential data. [51], [53]–[56].

d) xLSTM for audio & music: While xLSTMs have not been extensively utilized for long-term, structure-coherent music generation, their application in other audio and music-related studies has been demonstrated.In audio representation learning, AxLSTM [57] applies xLSTM to masked spectrogram modeling and outperforms SSAST baselines by up to $\approx 20\%$ on 10 downstream audio tasks with $\approx 45\%$ fewer parameters, indicating that xLSTM's recurrent-plus-associative memory can scale to real acoustic sequences and generalize across tasks. While AxLSTM targets representation learning (not generation), it supports the claim that xLSTM handles long, structured time series competitively.

C. The Evaluation Axis: Assessing Musical and Structural Quality

Evaluation remains one of the most complex and underdeveloped components of symbolic music generation research. Unlike text or image synthesis, music lacks a unified metric that simultaneously captures perceptual quality, stylistic fidelity, and structural coherence. Consequently, studies have employed a variety of evaluation strategies that can broadly be categorised into *quantitative*, *perceptual*, and *structure-aware* approaches [2], [10], [58].

a) Quantitative Metrics.: Early evaluations of neural music models followed conventions from language modeling, employing probabilistic metrics such as negative log-likelihood and perplexity to measure predictive accuracy. Additional statistical indicators including pitch-class histograms, rhythmic entropy, tonal distance, and n-gram similarity were later introduced to assess stylistic similarity between generated and training data [2], [10]. More recent benchmarking frameworks have sought to unify these measures within a reproducible evaluation pipeline. Liang et al. [59] introduced *MuSpike*, a multi-dimensional toolkit that evaluates symbolic music across pitch accuracy, rhythmic stability, tonal consistency, and structural diversity, integrating statistical metrics with perceptual similarity embeddings. While such frameworks enhance reproducibility and comparability, they remain primarily focused on local statistics and short-term correlations. Consequently, a model can achieve excellent likelihood or similarity scores while still producing compositions lacking thematic development or global form.

b) Perceptual and Listening-Based Evaluation.: Subjective listening studies are widely employed to complement quantitative analysis. Participants typically rate generated samples for musicality, creativity, realism, and structural coherence, providing insight into human perception of musical form. Chen et al. [58] emphasise that perceptual assessments capture expressive and affective qualities beyond the reach of numeric metrics. However, such evaluations are inherently subjective, labour-intensive, and often inconsistent across datasets and listener groups, limiting their reproducibility and comparability.

c) Structure-Aware and Segmentation-Based Metrics.: To address the lack of quantitative metrics in evaluating structural coherence, recent research has proposed structure-oriented evaluation frameworks that directly quantify musical form. De Berardinis et al. [5], [60] introduced the *Musical Structure Communities* (MSCOM) algorithm, which applies hierarchical community detection to self-similarity matrices to derive segmentations and compute structural complexity indices. These metrics capture repetition, contrast, and multi-level coherence which are core indicators of musical form, and allow direct comparison between generated and human-composed pieces.

Evaluation practices in symbolic music generation remain diverse and fragmented, with most studies relying on combinations of quantitative, perceptual, and structure-aware methods. While recent frameworks such as MSCOM have advanced the objective assessment of long-term coherence, these approaches are still not universally adopted across benchmarking studies. In line with prevailing practices and recent research, this work therefore plans to adopt a comprehensive evaluation strategy combining quantitative musical-quality measures, structural metrics, and human perceptual assessmentsto rigorously compare the proposed xLSTM model with established baselines.

IV. Synthesis, Research Gap, and Conclusion

Across the literature, two broad paradigms for modeling long-term structure in symbolic music emerge: *implicit* hierarchical models and *explicit* hierarchical models. Early implicit architectures such as conventional RNNs, LSTMs, and event-based Transformers are flexible and domain-agnostic, requiring no manual segmentation of musical form [35], [36], [61]. However, they consistently struggle to maintain coherence beyond short melodic spans, often losing thematic continuity over phrases and sections. This limitation is well documented in empirical evaluations of long-form symbolic generation [4], [61].

In response, the field shifted toward explicit hierarchical architectures such as MusicVAE, the Small-Tunes Transformer, bar-level Transformers and hierarchical diffusion models [8], [9], [45], [48]. These models demonstrate much stronger long-term coherence by organizing generation into fixed layers (notes $\rightarrow$ bars $\rightarrow$ phrases $\rightarrow$ sections). Yet, their strengths come at significant cost: they require hand-crafted segmentation, impose rigid structural templates, and generalize poorly

across genres or datasets with differing musical forms. As surveys note, the reliance on predefined structure limits scalability and reduces adaptability [10].

A. Research Gap

Taken together, the literature reveals a persistent gap: *no existing architecture simultaneously achieves the long-term structural modeling capability of explicit hierarchical designs while retaining the flexibility and generality of implicit sequence models*. Current approaches either impose manually defined hierarchical structure—limiting adaptability across musical styles—or rely on fully implicit mechanisms that struggle to maintain coherence over long time spans. This motivates the search for a scalable, domain-agnostic architecture capable of learning hierarchical musical dependencies directly from data, without predefined segmentation or rigid temporal levels.

Recent advances in recurrent modeling suggest a promising but unexplored direction. The extended Long Short-Term Memory (xLSTM) architecture introduces exponential gating and matrix memory, enabling *implicit multi-timescale dynamics* within a unified recurrent framework [50]. Studies in language modeling, long-context reasoning, and audio representation learning indicate that xLSTM can capture hierarchical temporal dependencies more effectively than standard LSTMs and, in some settings, rival Transformer-based models on long-range tasks [50], [53], [57].

Despite this alignment with the hierarchical nature of musical structure, **xLSTM has not yet been studied in the context of symbolic music generation**. In particular, it remains unknown whether its implicit multi-timescale memory mechanisms can reproduce—or potentially exceed—the long-term structural coherence achieved by explicitly hierarchical architectures.

B. Our Contribution

To address this gap, this work proposes to investigate whether xLSTM's implicit hierarchical dynamics can support long-term structural coherence in symbolic music generation, comparable to or exceeding that of explicit hierarchical and Transformer-based models. Specifically, this study aims to:

- Design and implement an xLSTM-based symbolic music generation model
- Benchmark xLSTM-based proposed model against representative explicit hierarchical baselines (MusicVAE [8], Museformer [41]) and strong Transformer variants (Music Transformer [14])
- Evaluate structural coherence using structure-aware metrics such as segmentation similarity, motif recurrence, and global form consistency, complemented by musicality measures and human listening evaluations [5].

Through this investigation, this work seeks to assess whether a modern implicit hierarchical architecture can narrow the longstanding trade-off between flexibility and structural control, and thereby contribute toward scalable, long-form symbolic music generation.

Acknowledgment

The authors would like to thank their supervisor, Dr. Isuru Nawinne, for his guidance and support throughout this research.

The authors also acknowledge the use of artificial intelligence–based tools during the preparation of this literature review. Specifically, *ChatGPT* and *Gemini* were used for language refinement, LaTeX code generation, and structural suggestions. In addition, *Consensus* and *SciSpace* were employed to assist with literature discovery and the summarization of research articles.

All factual content, citations, interpretations, and technical claims presented in this work were independently reviewed and verified by the authors, who take full responsibility for the accuracy and integrity of the final manuscript.

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